

WHERE PASSION MEETS PROPULSION



PRODUCT REQUIREMENTS DOCUMENT

AIRNOVA

Drone & Aerospace Committee — Official Website

Tagline: "Flying Beyond Limits"

Backdrop watermark: "Where passion meets propulsion"

Document Version	1.0
Prepared For	Airnova Drone & Aerospace Committee
Document Type	Website Product Requirements Document
Reference Benchmark	djsnova.space (structural reference)
Status	Draft for Review

1. Document Overview

This Product Requirements Document (PRD) defines the scope, structure, content, design language, and technical requirements for the official website of **Airnova**, the Drone & Aerospace committee. The website will serve as the committee's digital identity — communicating its vision, mission, subsystems, achievements, and projects to students, faculty, sponsors, and the wider drone/aerospace community. The structural benchmark for navigation and layout conventions is the existing club site **djsnova.space**, adapted to Airnova's aerospace/drone identity, tagline “**Flying Beyond Limits**”, and backdrop watermark “**Where passion meets propulsion.**”

The site must be a modern, fully **responsive** single-page-application style website (with smooth-scroll or routed sections) that works seamlessly across desktop, tablet, and mobile devices.

1.1 Purpose of this PRD

- Define the exact information architecture and section-by-section content for the Airnova website.
- Establish design, branding, and interaction requirements (including the watermark treatment).
- Specify technical, responsiveness, performance, and accessibility requirements for development.
- Provide a shared reference for designers, developers, and committee stakeholders to align on scope before build.
- Define success metrics and a phased rollout/timeline for implementation.

1.2 Goals & Objectives

Goal	Description	Priority
Brand presence	Establish Airnova as a credible, technically serious drone & aerospace committee through strong visual identity.	High
Recruitment	Attract prospective student members by showcasing subsystems, projects, and achievements.	High
Sponsor confidence	Present achievements, timeline, and technical depth to reassure and attract sponsors.	High
Showcase projects	Highlight flagship builds: gliders, VTOL, RC planes, rocketry, gesture-controlled drone, ornithopter, bionic butterfly.	High
Team storytelling	Introduce leadership (Captain) and subsystem teams to humanize the committee.	Medium
Responsiveness	Ensure flawless experience across all device sizes and orientations.	High

1.3 Target Audience

- **Prospective members** — students interested in drones, aerospace, robotics, and engineering.
- **Current members** — using the site as a reference point for subsystem info and achievements.
- **Sponsors & institutions** — evaluating credibility, achievements, and technical scope for funding/partnership.
- **Competition organizers & judges** — reviewing the committee's project portfolio and history.
- **College administration & faculty** — tracking committee activity and representation.

2. Brand & Design Language

2.1 Core Identity

Element	Specification
Committee Name	Airnova
Tagline	“Flying Beyond Limits” — displayed prominently in the intro/hero section.
Backdrop Watermark	“Where passion meets propulsion” — large, low-opacity, rotated or static typographic watermark behind hero/section backdrops (aircraft-carbon-fiber aesthetic).
Visual Theme	Aerospace / aviation-tech: deep navy & charcoal backgrounds, propulsion-orange or ion-blue accent, subtle grid/circuit or blueprint textures, drone/aircraft silhouettes.
Typography	Bold, modern sans-serif for headings (e.g., Orbitron/Space Grotesk/Poppins style); clean readable sans-serif for body text (e.g., Inter/Roboto).
Motion	Subtle parallax on hero, scroll-triggered fade/slide-ins, animated counters for achievements, drone-propeller spin micro-interactions on hover.

2.2 Color Palette (suggested)

Role	Color	Usage
Primary background	Deep Navy #0B1220	Base background across sections
Secondary background	Charcoal Slate #111A2E	Alternating section panels/cards
Accent	Propulsion Orange #FF7A1A	CTAs, highlights, hover states, icons
Text (muted)	Steel Blue-Grey #8FA3C4	Secondary text, captions, watermark
Text (primary)	Off-white #F5F7FB	Headings & body text on dark backgrounds

2.3 Watermark Implementation Requirement

The phrase **“Where passion meets propulsion”** must appear as a large-scale, low-opacity (approx. 4–8% opacity) typographic watermark in the backdrop of the hero/intro section, and optionally repeated subtly behind the Vision & Mission and Timeline sections. It should be implemented as a fixed or slow-parallax background layer (CSS background text, SVG, or canvas) that never interferes with foreground content legibility or accessibility contrast ratios.

3. Information Architecture & Sitemap

Following the reference site's convention of a persistent top navigation bar with a single-page scroll layout (plus dedicated sub-pages for deep content), Airnova's site will use the structure below.

Nav Item	Maps To Section	Type
Home	Intro / Hero	Anchor
About	Vision & Mission	Anchor
Subsystems	Our Subsystems	Anchor
Team	Leadership / Captain	Anchor / Sub-page
Journey	Timeline	Anchor
Projects	Flagship Projects & Gallery	Anchor / Sub-page
Achievements	Awards & Competitions	Sub-page (like reference)
Sponsors	Sponsorship & Partners	Anchor
Contact	Footer / Contact Section	Anchor

3.1 Full Page Flow (Top to Bottom)

- 1. Sticky Navigation Bar (logo + links + CTA button e.g. "Join Airnova" / "Recruitment").
- 2. Hero / Intro Section — full-viewport, tagline, animated drone/aircraft visual, watermark backdrop.
- 3. Vision & Mission Section — two-column or card layout.
- 4. Our Subsystems Section — grid of 8 subsystem cards.
- 5. Leadership / Captain Section — photo + bio placeholder card.
- 6. Timeline Section — 2019 → 2021 → 2023 → 2025 milestone journey.
- 7. Flagship Projects Section — gliders, VTOL, RC planes, rocketry, gesture-controlled drone, ornithopter, bionic butterfly.
- 8. Achievements / Gallery Section — competition wins, photos, press mentions (reference: Astrophotography/Achievements pages).
- 9. Sponsorship Section — sponsor tiers, logos, "Partner with us" CTA.
- 10. Team / Members Grid (optional expansion) — subsystem-wise member cards.
- 11. Footer — quick links, social icons, contact/location, copyright.

4. Section-by-Section Requirements

4.1 Intro / Hero Section

Purpose: First impression — immediately communicate identity, tagline, and energy of the committee.

- Full-width, full (or near-full) viewport height section with a dark aerospace-themed background.
- Large heading: **“AIRNOVA”** with sub-heading tagline **“Flying Beyond Limits.”**
- Background: subtle animated drone/aircraft graphic, particle or cloud/sky motion, or blueprint-grid texture.
- Backdrop watermark text “Where passion meets propulsion” rendered large and faint behind the content.
- Primary CTA button (e.g., “Explore Our Work” / “Join the Squadron”) scrolling to Subsystems or About.
- Secondary CTA or scroll-indicator (animated chevron) prompting scroll to next section.
- Fully responsive: stacks vertically on mobile, scales typography with viewport (clamp/rem units).

4.2 Vision & Mission Section

Purpose: Communicate the committee's guiding purpose, mirroring the reference site's “Our Mission” block but expanded into a two-part Vision + Mission layout.

Field	Content
Mission	Advance Drone & Aerospace Tech
Vision	Pioneer. Build. Innovate.

- Displayed as two adjacent cards or a split-panel layout (Vision left, Mission right — or stacked on mobile).
- Each card includes an icon (compass/rocket for Vision, propeller/circuit for Mission).
- Supporting one-line elaboration text may be added under each (placeholder copy to be finalized by committee).
- Scroll-triggered fade/slide-up animation on entry.

4.3 Our Subsystems Section

Purpose: Showcase the 8 functional subsystems that structure the committee, similar to a departments/teams grid.

#	Subsystem	Suggested Icon
1	Aerodynamics & Structure	Wing/airfoil icon
2	Propulsion	Rocket/thruster icon
3	Control Systems	Circuit board/joystick icon
4	Research & Development	Microscope/flask icon
5	Sponsorship	Handshake icon
6	Documentation	Document/clipboard icon
7	Finance & Inventory	Ledger/wallet icon
8	Social Media	Megaphone/at-symbol icon

- Responsive grid: 4 columns (desktop) → 2 columns (tablet) → 1 column (mobile).
- Each card: icon, subsystem name, 1–2 line description (placeholder), subtle hover-lift with orange glow border.
- Optional: clicking a card expands to show subsystem lead(s) and members once data is available.

4.4 Leadership — Captain Section

Purpose: Humanize the committee by introducing its Captain, matching the reference site's team-highlight convention.

- Layout: photo on one side (circular or hexagonal frame with orange accent ring), name/title and description on the other.
- Fields required: **Full Name**, **Title** ("Captain, Airnova"), **Photo**, **Bio/Description** (2–4 sentences).
- **Content status:** Photo and description are placeholders to be finalized and added later by the committee — implement as a clearly structured, easily editable content block (e.g., CMS field or config object) so it can be updated without a code change.
- Include social/contact icons (LinkedIn, Instagram, email) for the Captain, optional.
- Design should reserve space to later expand into a full leadership grid (Vice-Captain, subsystem leads) without breaking layout.

4.5 Timeline / Our Journey Section

Purpose: Show the committee's growth story as a vertical (mobile) / horizontal or vertical (desktop) animated timeline.

Year	Milestone
2019	Airnova founded — the committee's inception.
2021	Expanded portfolio — growth in subsystems, membership, and technical scope.
2023	Major achievements — wins at national and international events and competitions.
2025	First flagship projects delivered: gliders, VTOL, RC planes, rocketry, gesture-controlled drone, ornithopter, bionic butterfly.

- Visual: connected vertical line/rail with milestone nodes (dot/icon) and alternating left-right content cards on desktop; single-column stacked on mobile.
- Scroll-driven progressive reveal (line "draws" itself and nodes light up as the user scrolls).
- Each node expandable/clickable for more detail if content grows in future.

4.6 Flagship Projects Section

Purpose: Showcase Airnova's engineering output — the 2025 project cohort — as a visually rich card gallery or carousel.

Project	Category
Gliders	Fixed-wing / Aerodynamics
VTOL	Vertical Take-Off & Landing aircraft
RC Planes	Radio-controlled fixed-wing aircraft
Rocketry	Propulsion / Aerospace research
Gesture-Controlled Drone	Control Systems / HCI innovation
Ornithopter	Bio-inspired flapping-wing flight
Bionic Butterfly	Bio-mimetic micro air vehicle

- Each project card: image/render placeholder, project name, category tag, 1–2 line description, “Learn More” link (optional detail modal or sub-page).
- Filterable/tab layout by category (Fixed-Wing, VTOL, Bio-Inspired, Propulsion, Control Systems) — nice-to-have, not required for v1.
- Responsive grid or horizontally swipeable carousel on mobile.

4.7 Achievements & Gallery Section

Purpose: Mirrors the reference site's dedicated Achievements/Astrophotography pages — build credibility through visual proof of wins and activity.

- Achievements sub-page or section listing competition wins (2023 national/international wins referenced in Timeline) with event name, year, and result — content to be finalized by committee.
- Photo gallery grid of events, workshops, and builds (masonry or grid layout, lightbox on click).
- Optional animated counters (e.g., “X Competitions,” “Y Projects Built,” “Z Members”) for quick-glance credibility.

4.8 Sponsorship Section

Purpose: Support the Sponsorship subsystem's goals by giving sponsors a clear, professional showcase and call-to-action.

- Sponsor logo strip/grid (current and past sponsors — placeholder until provided).
- Sponsorship tiers or “Why Partner with Airnova” value proposition block.
- CTA button linking to a contact form or email for sponsorship inquiries.

4.9 Footer / Contact Section

Purpose: Provide navigation, social presence, and contact information, matching the reference site's footer structure.

- Committee logo + one-line description (“Airnova — the Drone & Aerospace committee of [College Name].”).
- Social icons: Instagram, LinkedIn, X/Twitter, YouTube (for build/flight demo videos).
- Quick links column mirroring nav (Home, About, Subsystems, Projects, Achievements, Contact).
- Contact details: college address (map link), committee email, point-of-contact phone number.
- Copyright line with current year, auto-updated via script.

5. Interaction, Responsiveness & Accessibility

5.1 Responsive Breakpoints

Breakpoint	Width Range	Layout Behavior
Mobile	< 640px	Single column, hamburger nav, stacked cards, larger tap targets (min 44px).
Tablet	640px – 1024px	2-column grids, condensed nav, timeline remains vertical.
Desktop	1024px – 1440px	Full multi-column grids (3–4 cols), horizontal nav, side-by-side Vision/Mission.
Large Desktop	> 1440px	Max content width container (e.g., 1280–1440px) centered, generous whitespace.

5.2 Reactive / Interaction Requirements

- Site must be built with a component-based reactive framework (e.g., React, Next.js, or Vue) enabling dynamic state-driven UI updates — matching the reference site's Next.js foundation.
- Sticky/transforming navbar (transparent on hero, solid on scroll).
- Scroll-triggered entrance animations for each section (fade/slide) using a library such as Framer Motion / GSAP / AOS.
- Interactive hover states on all cards, buttons, and nav links with smooth transitions (200–300ms ease).
- Mobile hamburger menu with slide-in/overlay navigation drawer.
- Image lazy-loading and next-gen formats (WebP/AVIF) for performance.
- Smooth-scroll anchor navigation between sections.
- Loading states/skeletons for any dynamically fetched content (e.g., achievements pulled from a CMS).

5.3 Accessibility (A11y)

- Maintain WCAG 2.1 AA color contrast, especially for text over the watermark/dark backgrounds.
- All interactive elements keyboard-navigable and screen-reader labeled (alt text, aria-labels).
- Respect prefers-reduced-motion for users who disable animations.
- Semantic HTML structure (header, nav, main, section, footer) for SEO and assistive tech.

6. Technical Requirements

6.1 Suggested Tech Stack

Layer	Recommendation
Framework	Next.js (React) — matches reference site's stack; enables SSR/SSG for fast loads and SEO.
Styling	Tailwind CSS for utility-based responsive styling; CSS modules for custom watermark/animation effects.
Animation	Framer Motion (scroll/entry animations), optionally GSAP for the timeline's scroll-driven line-draw effect.
Icons	Lucide/Heroicons for UI icons; custom SVGs for subsystem/drone-specific icons.
Content Management	Structured JSON/Markdown config (or headless CMS) for team, timeline, and project data — enables easy future edits (e.g., Captain photo/bio).
Hosting	Vercel / Netlify (static/SSG hosting with CI/CD from GitHub).
Forms	Formspree / EmailJS / custom backend for sponsorship & contact form submissions.

6.2 Performance Requirements

- Lighthouse Performance score ≥ 90 on mobile and desktop.
- First Contentful Paint (FCP) under 1.8s on 4G; Largest Contentful Paint (LCP) under 2.5s.
- Total page weight optimized via compressed images, code-splitting, and lazy-loaded below-fold sections.
- Cumulative Layout Shift (CLS) near zero — reserve space for images/watermark elements to avoid layout jumps.

6.3 SEO & Metadata

- Descriptive meta title/description per page (e.g., “Airnova — Drone & Aerospace Committee | Flying Beyond Limits”).
- Open Graph & Twitter Card tags for social sharing previews (logo + tagline image).
- Semantic heading hierarchy (single H1 per page) and descriptive alt text for all images.
- sitemap.xml and robots.txt included; favicon and app icons for all device sizes.

6.4 Browser & Device Support

- Latest two versions of Chrome, Safari, Firefox, and Edge.
- iOS Safari and Android Chrome (mobile) fully supported.
- Graceful degradation of advanced animations on low-power/older devices.

7. Content Inventory & Placeholder Tracker

The table below tracks which content is finalized versus pending, so the development team can build with placeholders where noted and swap in final assets later without structural rework.

Section	Content Status	Notes
Tagline & Watermark	Final	"Flying Beyond Limits" / "Where passion meets propulsion."
Vision & Mission	Final	As provided by committee.
Subsystems (8)	Final (names)	Descriptions/icons to be designed; names confirmed.
Captain Photo & Bio	Pending	Build with placeholder photo/text block; easily editable field.
Timeline Milestones	Final (years/summary)	2019, 2021, 2023, 2025 confirmed; can add more detail later.
Flagship Projects (7)	Final (names)	Images/descriptions pending from R&D; subsystem.
Achievements list	Pending	Specific event names/years to be supplied by committee.
Sponsor logos	Pending	Build sponsor grid to be populated later.
Contact details	Pending	College address, email, phone to be confirmed.

8. Success Metrics (KPIs)

Metric	Target
Mobile usability / responsiveness	Zero layout-breaking issues across breakpoints in QA.
Page load performance	Lighthouse score ≥ 90 (Performance, Accessibility, SEO).
Recruitment form submissions	Measurable increase post-launch vs. prior recruitment cycle.
Sponsor inquiries	Track via sponsorship CTA form submissions.
Average session duration	Increase engagement time via analytics (target > 1.5 min).
Bounce rate	Below 55% on landing/hero section.

9. Development Roadmap (Suggested Phases)

Phase	Scope	Est. Duration
Phase 1 — Design	Wireframes, high-fidelity mockups (Figma), brand asset finalization, watermark treatment.	1–2 weeks
Phase 2 — Core Build	Hero, Vision/Mission, Subsystems, Timeline, Footer — static responsive build.	2–3 weeks
Phase 3 — Rich Content	Projects gallery, Achievements page, Captain/Team section, Sponsorship section.	1–2 weeks
Phase 4 — Animation & Polish	Scroll animations, micro-interactions, watermark parallax, performance tuning.	1 week
Phase 5 — QA & Launch	Cross-device/browser testing, accessibility audit, SEO setup, deployment.	3–5 days

10. Open Questions / Pending Decisions

- Captain's name, photo, and bio copy — pending from committee.
- Specific achievement/competition names and years for the Achievements section.
- Final sponsor list and logo assets.
- College name and official contact details for the footer.
- Whether individual sub-pages (e.g., dedicated Achievements or Projects pages like the reference site) are required for v1, or if a single-page scroll site suffices initially.

— End of Document —